



### Greener recovery plans

The authors of the paper call for greener plans for recovery from the pandemic.  
<http://digit.in/aug21-58>



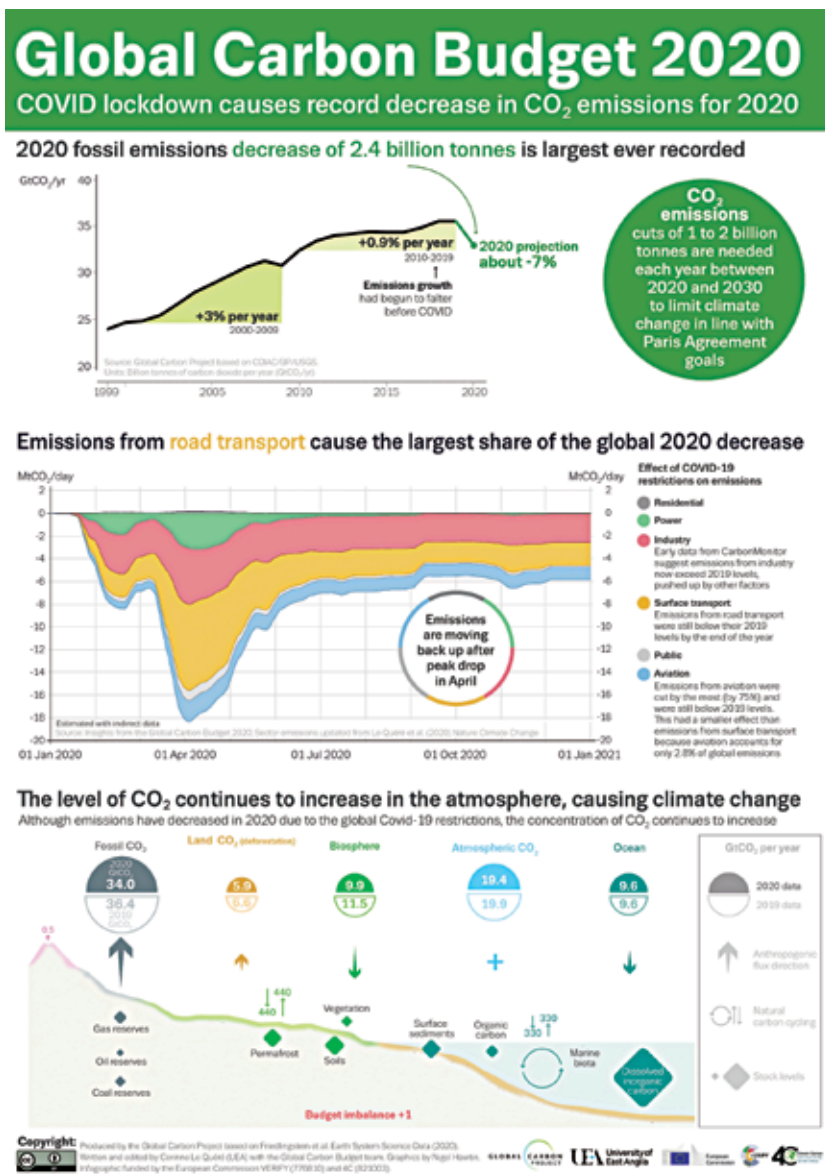
### Mitigating pathways

A report on all the different ways we can move towards zero net carbon emissions in time. <http://digit.in/aug21-59>

more complex and intricate tool to study the relationship between the rising temperature of the Earth and the amount of carbon dioxide gas causing it. And as we told earlier, since there are many organizations and study groups all over the world compiling their own independent reports, it is not that easy to give out a definite formula. This is a general idea of how research organisations arrive at the number.

The carbon budget is a relation between the carbon dioxide saturation levels in the atmosphere and the rise in temperature of the Earth. So, the first step in determining the carbon budget was to know what temperature rise will affect the Earth severely. In The Paris Agreement, this was specified to be 1.5 degrees Celsius. The next step is to figure out exactly how much carbon dioxide is emitted. Scientific research groups working both for government and private entities conduct surveys to investigate how much carbon dioxide is being released where and why.

One of the most sophisticated estimates so far is in a paper called "An integrated approach to quantifying uncertainties in the remaining carbon budget", published in January this year in Nature, an estimated 440 billion tons of carbon dioxide, in total, is the available budget for emissions after 2020. Even if we are able to restrict the amount of carbon dioxide emissions below this number, we will still only have a 50 per cent chance of meeting the 1.5-degree Celsius goal by the year 2040. Additionally, even this coin flip requires that by 2040, global carbon dioxide emissions should reach net zero, which means to keep disaster at bay we have to start sequestering the carbon. In order to increase this probability of meeting the 1.5 degrees Celsius, we will have to restrict the emissions to 230 billion tons. If things go ahead in a business as usual scenario, we will exhaust the entire available carbon budget completely within a period of ten years.



The atmospheric buildup of carbon does not halt even with drastic emissions reduction

### ARE WE DOOMED?

It is not all doom and gloom. Knowing the goals to aim for and agreeing on them really helps drive action. The last couple of years have seen a slight fall in the amount of carbon dioxide being released into the atmosphere. In 2019, as the numbers suggest, the increase in the amount of carbon dioxide released was only 0.1% above that of 2018. In 2020, because of the pandemic there was a record 7 per cent drop in the emission levels. If all the people on the planet manage to reduce emissions at the same rate, then we can go down to

net zero carbon emissions by 2035. The biggest reduction in emissions was caused by a decrease in road and air travel. While the ability to move out and about will return to individuals as we move into a post pandemic future, hopefully the rapid digitisation of technology combined with increased familiarity to remote working and collaboration might help push humans to reducing the amount of travel needed for work, which in turn would be beneficial to the planet. This is what is needed to keep disaster at bay, and meeting the goals that we have set for ourselves. **d**